LTSpice Notch Quality Report

test hoy

Auto-detected swept parameters and quality-factor (Q = f0 / BW @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 1

Total simulation steps : 40

`V(act+,act-)`

swept parameters: Cp, Itail

Report contents (per signal column):

1. All curves — Bode plot (magnitude + phase)
2. Magnitude grouped by each swept parameter (one subplot per fixed value)
3. Phase grouped by each swept parameter (one subplot per fixed value)
4. Quality-factor table for that column

End of report (extra analyses):

A. Cross-signal comparison (act vs pas, if both present)

B. Q vs each swept parameter (one line per other-param combo)

C. BW vs each swept parameter

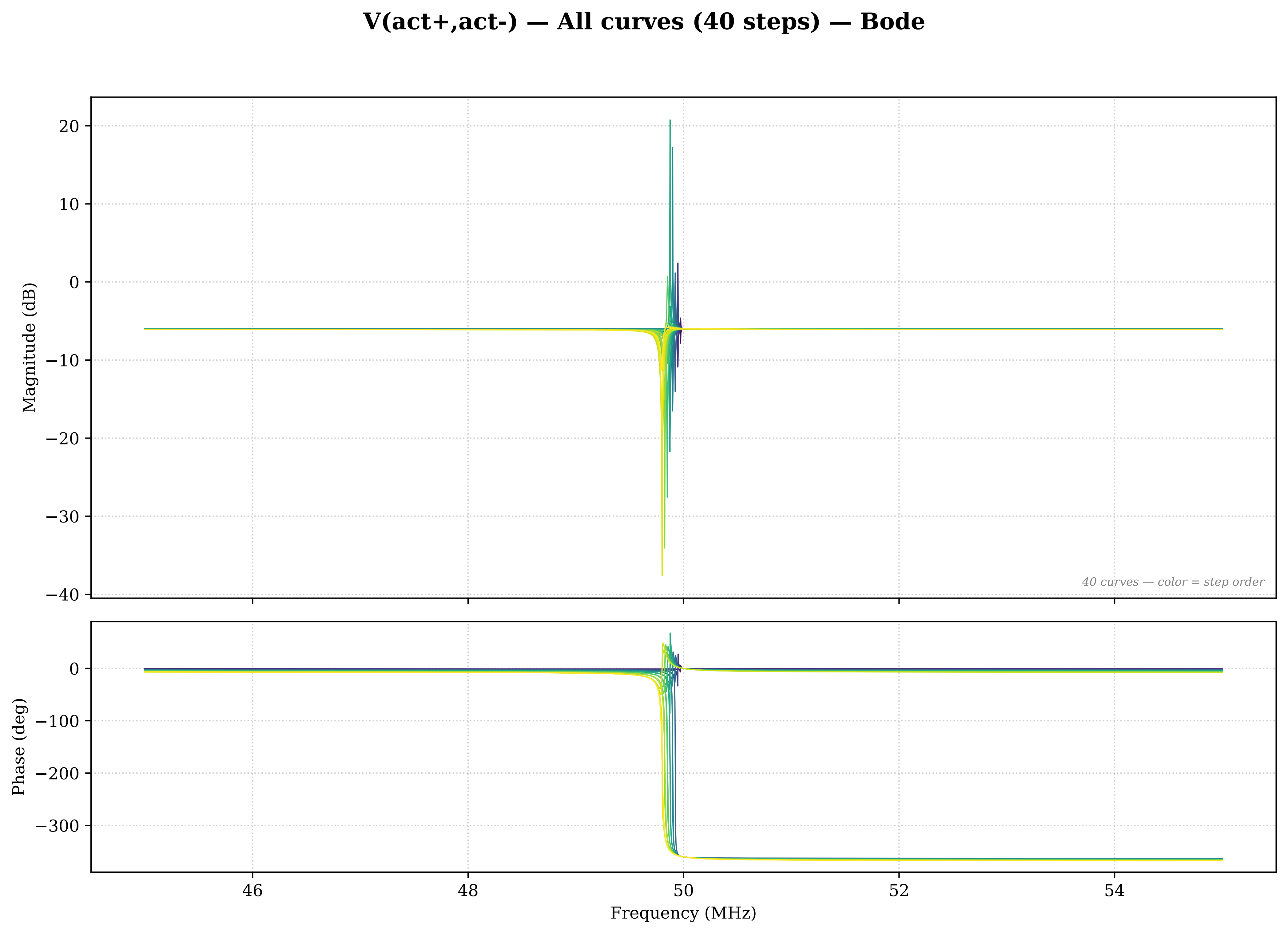
D. Q heatmap for V(act): rows × columns swept params, colored

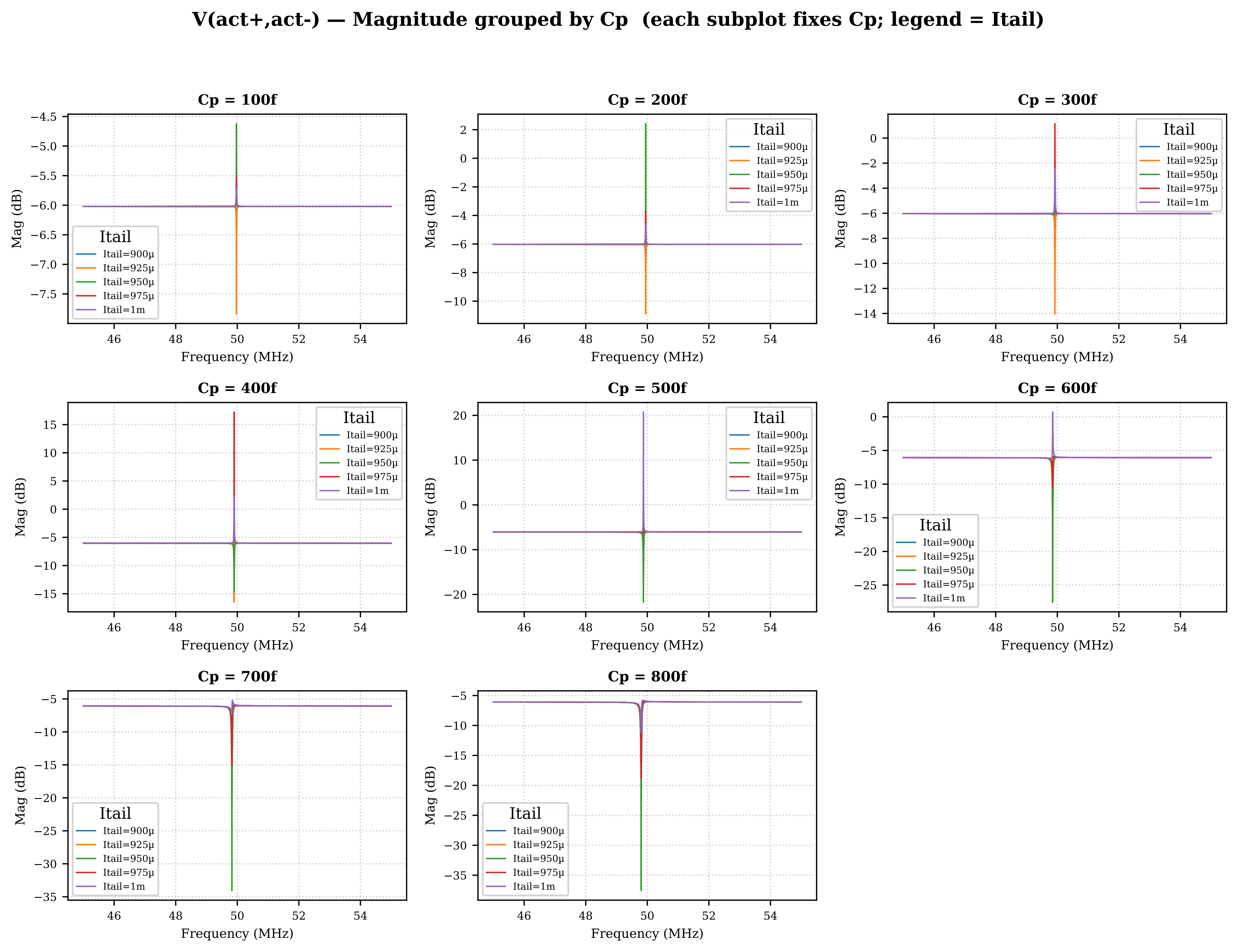
E. Combined quality-factor table across all signals

Signal: V(act+,act-)

Swept parameters: Cp, Itail

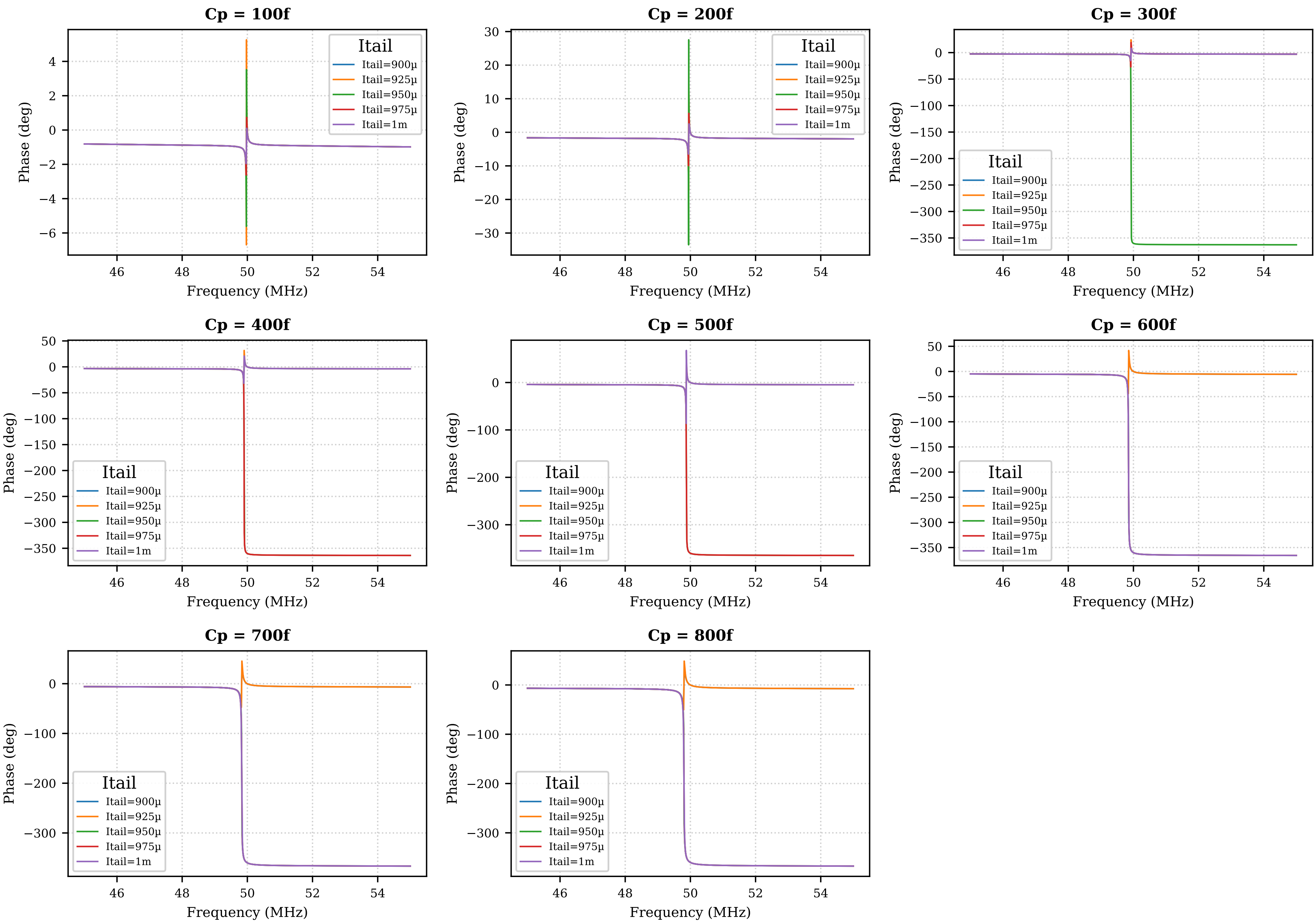
Steps: 40


V(act+,act-) — All curves (40 steps) — Bode
Magnitude (dB)
Phase (deg)
Frequency (MHz)
40 curves — color = step order
20
10
0
−10
−20
−30
−40
−100
−200
−300
46
48
50
52
54

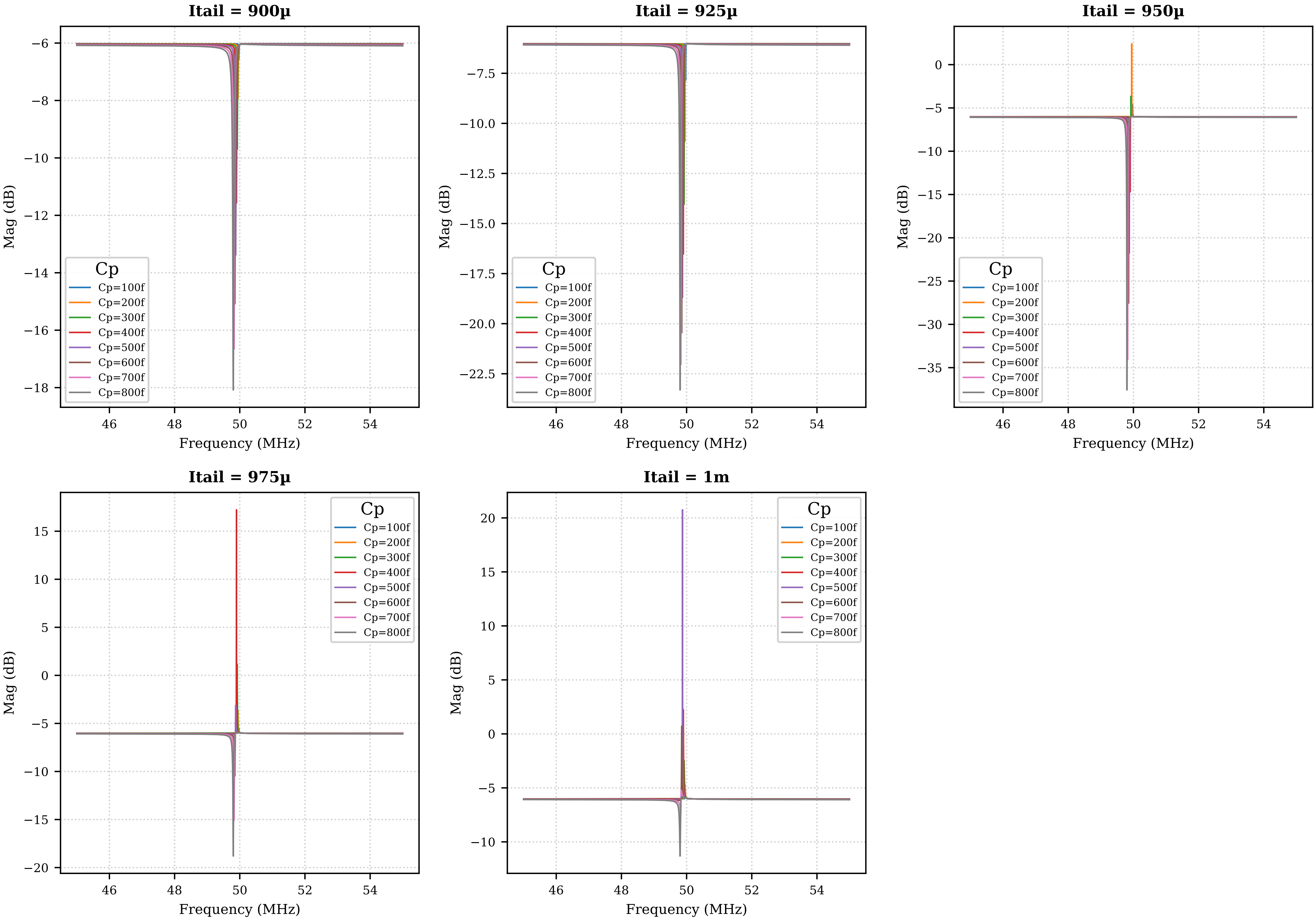


V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)
Cp = 100f
Cp = 200f
Cp = 300f
Cp = 400f
Cp = 500f
Cp = 600f
Cp = 700f
Cp = 800f
Mag (dB)
Frequency (MHz)
Itail
Itail=900µ
Itail=925µ
Itail=950µ
Itail=975µ
Itail=1m

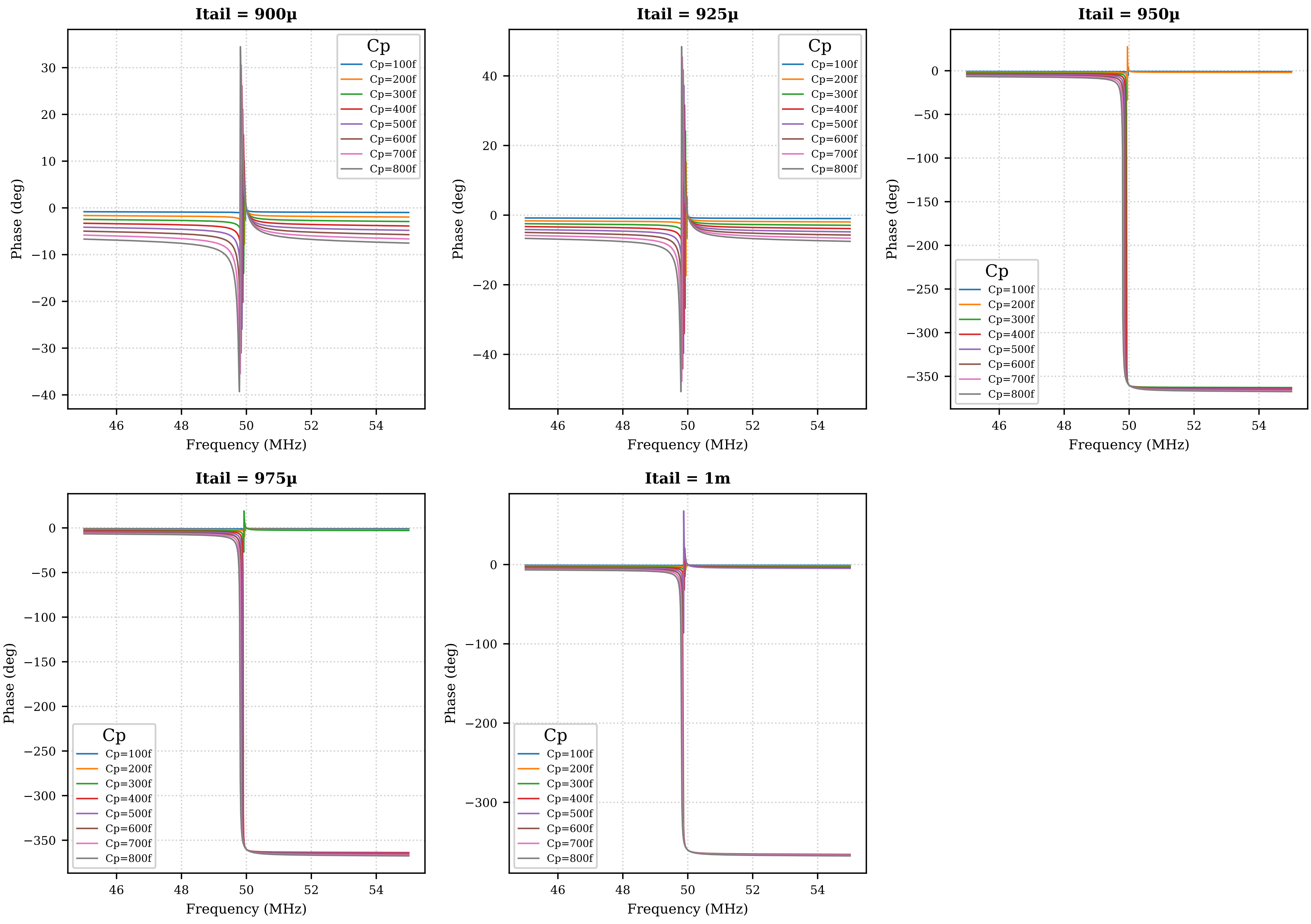
V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)



V(act+,act-) — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)



V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)



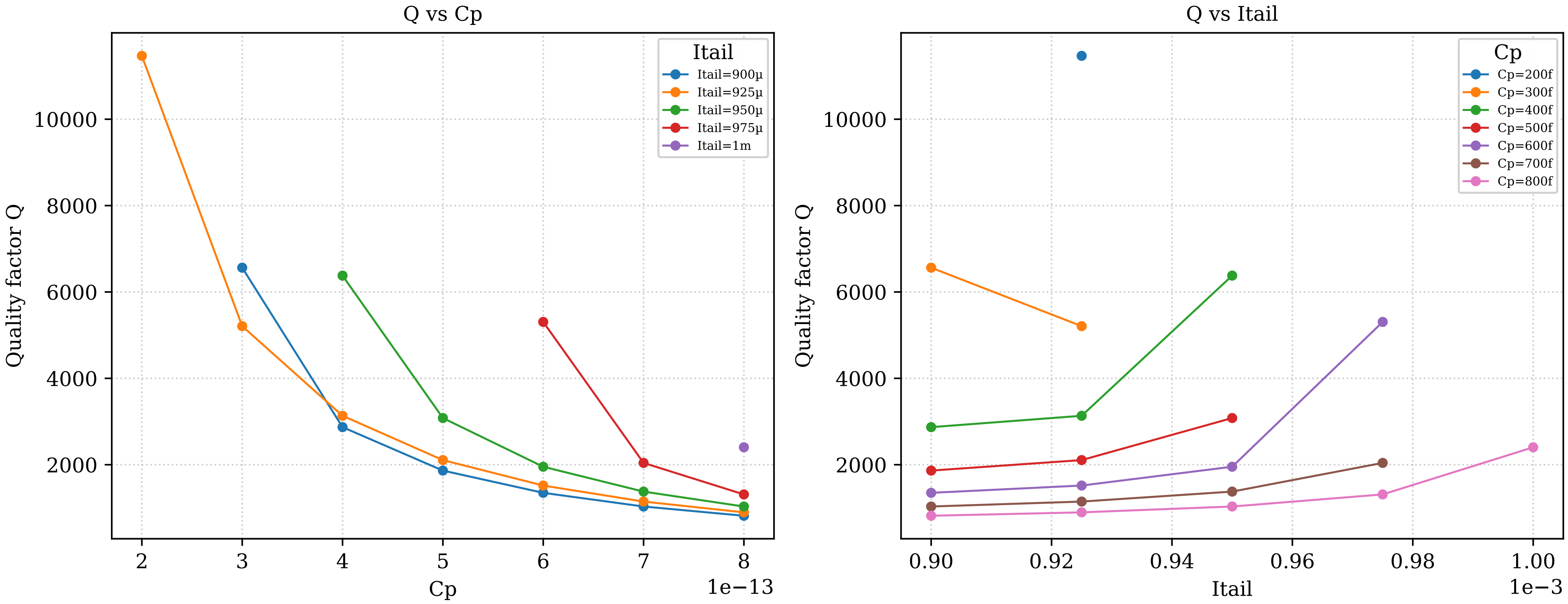
V(act+,act-) — Quality factor table (page 1/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	900µ	49.97	-6.022	0.5453			notch shallower than 3 d
V(act+,act-)	100f	925µ	49.97	-6.022	1.817			notch shallower than 3 d
V(act+,act-)	100f	950µ	49.81	-6.022	0.0002708			notch shallower than 3 d
V(act+,act-)	100f	975µ	55	-6.022	0.0001987			notch shallower than 3 d
V(act+,act-)	100f	1m	55	-6.022	0.0001989			notch shallower than 3 d
V(act+,act-)	200f	900µ	49.95	-6.025	1.903			notch shallower than 3 d
V(act+,act-)	200f	925µ	49.95	-6.025	4.87	4356	1.147e+04	
V(act+,act-)	200f	950µ	49.88	-6.025	0.004292			notch shallower than 3 d
V(act+,act-)	200f	975µ	49.7	-6.025	0.001084			notch shallower than 3 d
V(act+,act-)	200f	1m	55	-6.025	0.0007507			notch shallower than 3 d
V(act+,act-)	300f	900µ	49.92	-6.03	3.668	7607	6563	
V(act+,act-)	300f	925µ	49.92	-6.03	8.009	9585	5208	
V(act+,act-)	300f	950µ	49.91	-6.03	0.1097			notch shallower than 3 d
V(act+,act-)	300f	975µ	49.78	-6.03	0.005968			notch shallower than 3 d
V(act+,act-)	300f	1m	49.64	-6.03	0.002996			notch shallower than 3 d
V(act+,act-)	400f	900µ	49.9	-6.038	5.528	1.739e+04	2870	
V(act+,act-)	400f	925µ	49.9	-6.038	10.49	1.593e+04	3133	
V(act+,act-)	400f	950µ	49.9	-6.038	8.658	7823	6379	
V(act+,act-)	400f	975µ	49.82	-6.038	0.02709			notch shallower than 3 d
V(act+,act-)	400f	1m	49.72	-6.038	0.01076			notch shallower than 3 d
V(act+,act-)	500f	900µ	49.87	-6.047	7.334	2.675e+04	1864	
V(act+,act-)	500f	925µ	49.87	-6.047	12.64	2.367e+04	2107	
V(act+,act-)	500f	950µ	49.87	-6.047	15.71	1.619e+04	3081	
V(act+,act-)	500f	975µ	49.86	-6.047	0.2116			notch shallower than 3 d
V(act+,act-)	500f	1m	49.77	-6.047	0.03478			notch shallower than 3 d
V(act+,act-)	600f	900µ	49.85	-6.059	9.026	3.695e+04	1349	
V(act+,act-)	600f	925µ	49.85	-6.059	14.39	3.284e+04	1518	
V(act+,act-)	600f	950µ	49.85	-6.059	21.5	2.552e+04	1953	
V(act+,act-)	600f	975µ	49.85	-6.059	4.41	9394	5306	
V(act+,act-)	600f	1m	49.8	-6.059	0.1318			notch shallower than 3 d
V(act+,act-)	700f	900µ	49.82	-6.072	10.58	4.827e+04	1032	
V(act+,act-)	700f	925µ	49.82	-6.072	15.96	4.345e+04	1147	

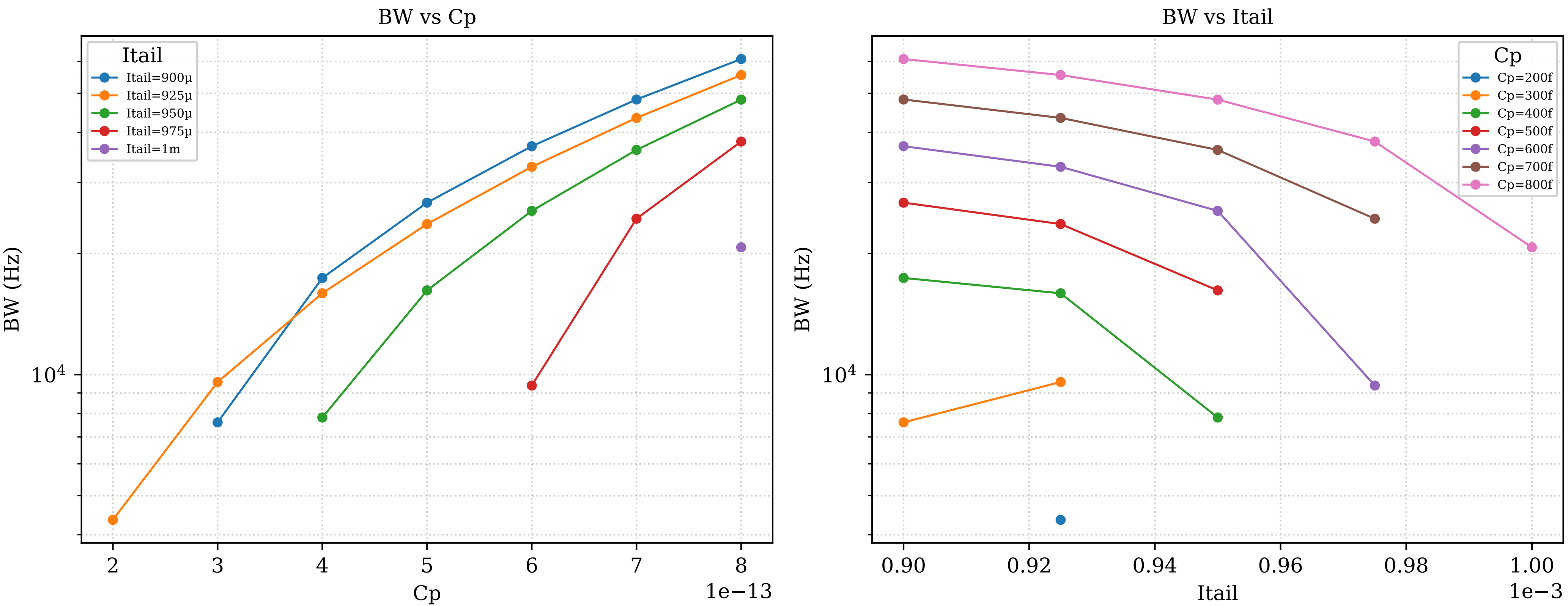
V(act+,act-) — Quality factor table (page 2/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	700f	950µ	49.82	-6.072	28	3.617e+04	1377	
V(act+,act-)	700f	975µ	49.82	-6.072	9.01	2.44e+04	2042	
V(act+,act-)	700f	1m	49.82	-6.072	1.314			notch shallower than 3 d
V(act+,act-)	800f	900µ	49.8	-6.088	11.99	6.086e+04	818.3	
V(act+,act-)	800f	925µ	49.8	-6.088	17.22	5.552e+04	897	
V(act+,act-)	800f	950µ	49.8	-6.088	31.49	4.824e+04	1032	
V(act+,act-)	800f	975µ	49.8	-6.088	12.71	3.795e+04	1312	
V(act+,act-)	800f	1m	49.8	-6.088	5.225	2.072e+04	2404	

B. V(act+,act-) — Quality factor vs swept parameters



C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=900µ	Itail=925µ	Itail=950µ	Itail=975µ	Itail=1m
Cp=100f	no BW	no BW	no BW	no BW	no BW
Cp=200f	no BW	11467	no BW	no BW	no BW
Cp=300f	6563	5208	no BW	no BW	no BW
Cp=400f	2870	3133	6379	no BW	no BW
Cp=500f	1864	2107	3081	no BW	no BW
Cp=600f	1349	1518	1953	5306	no BW
Cp=700f	1032	1147	1377	2042	no BW
Cp=800f	818	897	1032	1312	2404

Red = BW could not be measured | Yellow = Q computed ($Q \leq 1400$) | Green = $Q > 1400$

E. Combined quality-factor table — all signals (page 1/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	900µ	49.97	-6.022	0.5453			notch shallower than 3 d
V(act+,act-)	100f	925µ	49.97	-6.022	1.817			notch shallower than 3 d
V(act+,act-)	100f	950µ	49.81	-6.022	0.0002708			notch shallower than 3 d
V(act+,act-)	100f	975µ	55	-6.022	0.0001987			notch shallower than 3 d
V(act+,act-)	100f	1m	55	-6.022	0.0001989			notch shallower than 3 d
V(act+,act-)	200f	900µ	49.95	-6.025	1.903			notch shallower than 3 d
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V(act+,act-)	300f	950µ	49.91	-6.03	0.1097			notch shallower than 3 d
V(act+,act-)	300f	975µ	49.78	-6.03	0.005968			notch shallower than 3 d
V(act+,act-)	300f	1m	49.64	-6.03	0.002996			notch shallower than 3 d
V(act+,act-)	400f	900µ	49.9	-6.038	5.528	1.739e+04	2870	
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V(act+,act-)	700f	900µ	49.82	-6.072	10.58	4.827e+04	1032	
V(act+,act-)	700f	925µ	49.82	-6.072	15.96	4.345e+04	1147	

E. Combined quality-factor table — all signals (page 2/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	700f	950µ	49.82	-6.072	28	3.617e+04	1377	
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